



SERIES REACTIVE CIRCUITS

AC CIRCUITS

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

Phasor Diagram

Series R-L Circuit

Series R-C Circuit

Series R-L-C Circuit

Power Factor



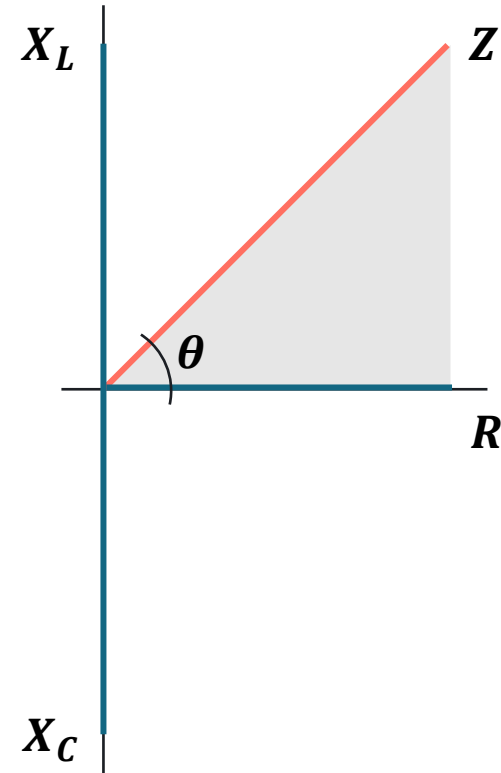
PHASOR DIAGRAM



PHASOR DIAGRAM

Phasor diagram is a graphical representation of magnitude and phase relationship between sinusoidal quantities.

Phasor Diagram



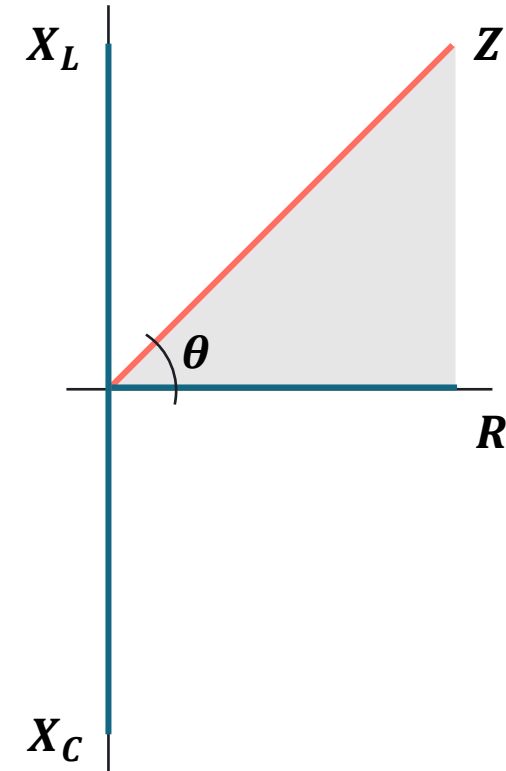
IMPEDANCE

Phasor Diagram

The impedance (Z) represents the total opposition offered by circuit elements (including resistance and reactance) to the flow of alternating current (AC).

Formula

$$Z = R + j(X_L - X_C)$$



INDUCTIVE REACTANCE

Phasor Diagram

The inductive reactance (X_L) represents the opposition offered by the inductor to the flow of alternating current (AC).

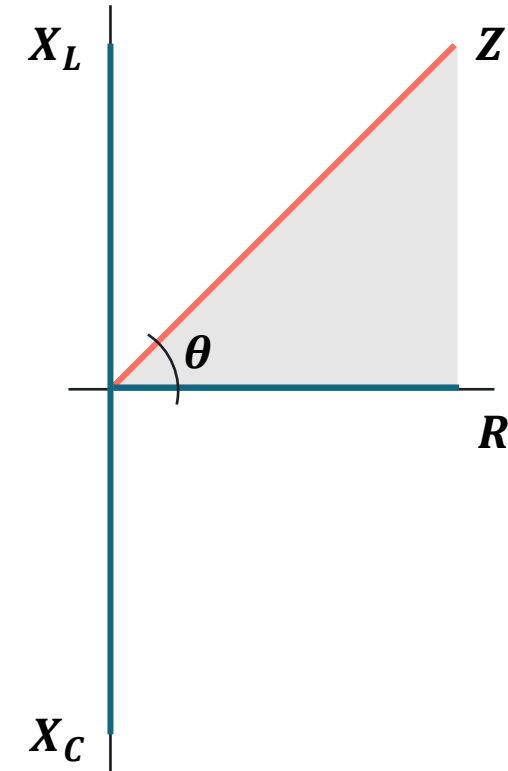
Formula

$$X_L = 2\pi fL$$

where:

f = frequency (Hz)

L = inductance (H)



CAPACITIVE REACTANCE

Phasor Diagram

The capacitive reactance (X_C) represents the opposition offered by the capacitor to the flow of alternating current (AC).

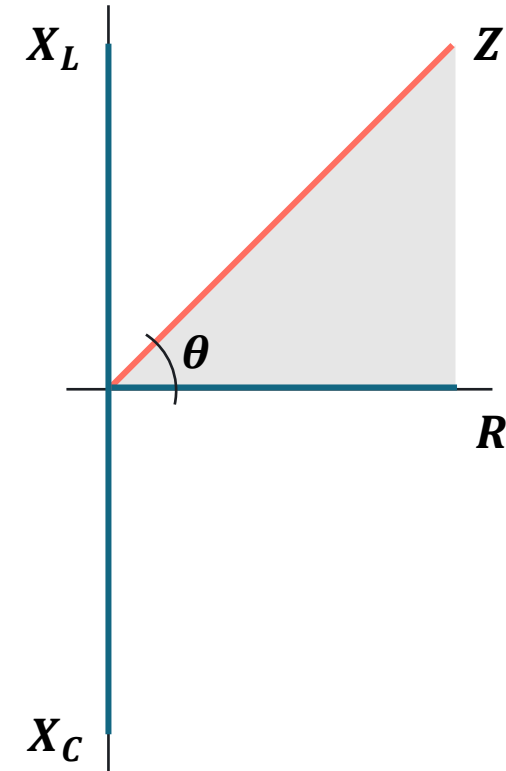
Formula

$$X_C = \frac{1}{2\pi f C}$$

where:

f = frequency (Hz)

C = capacitance (F)

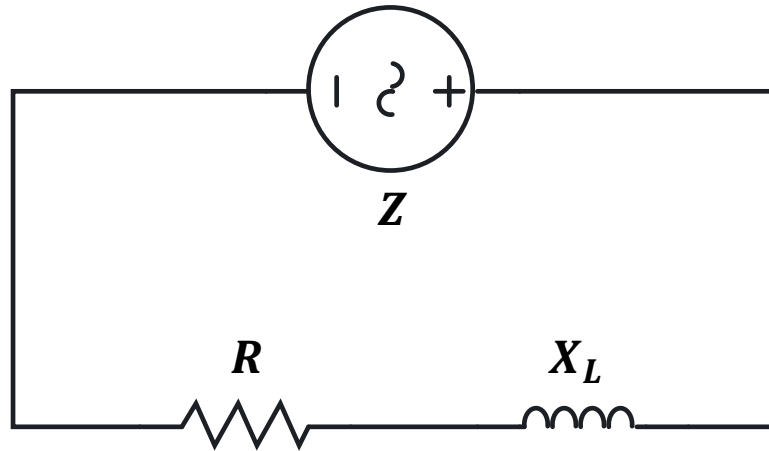


SERIES R-L CIRCUIT



SERIES R-L CIRCUIT

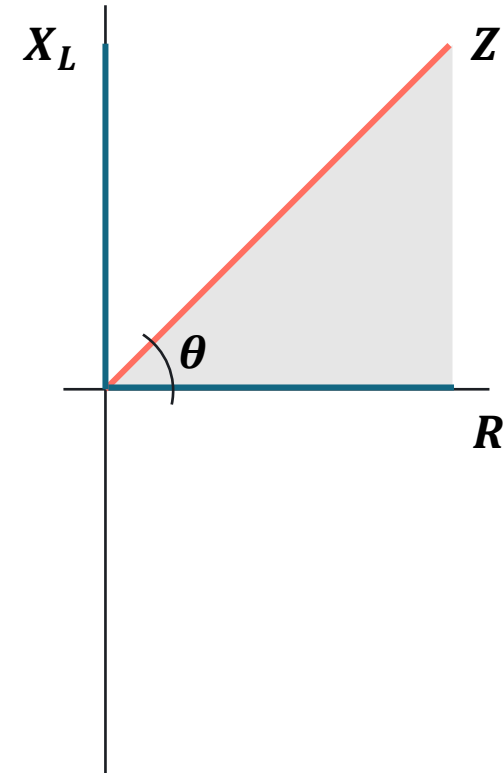
Circuit Diagram



Formula

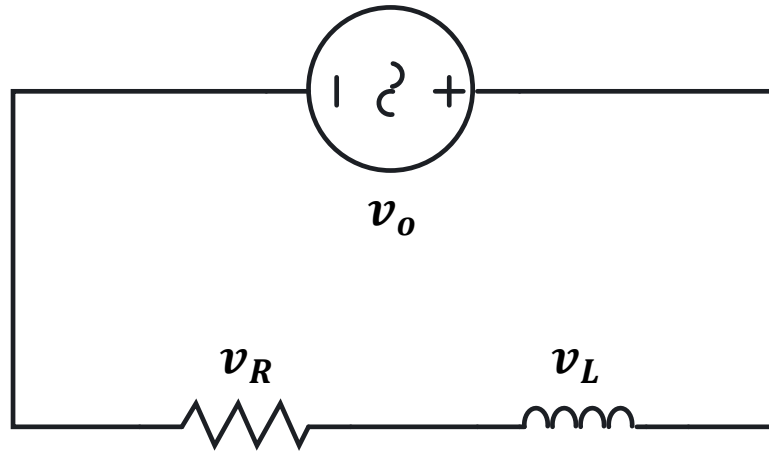
$$Z = R + jX_L$$

Impedance Phasor Diagram



SERIES R-L CIRCUIT

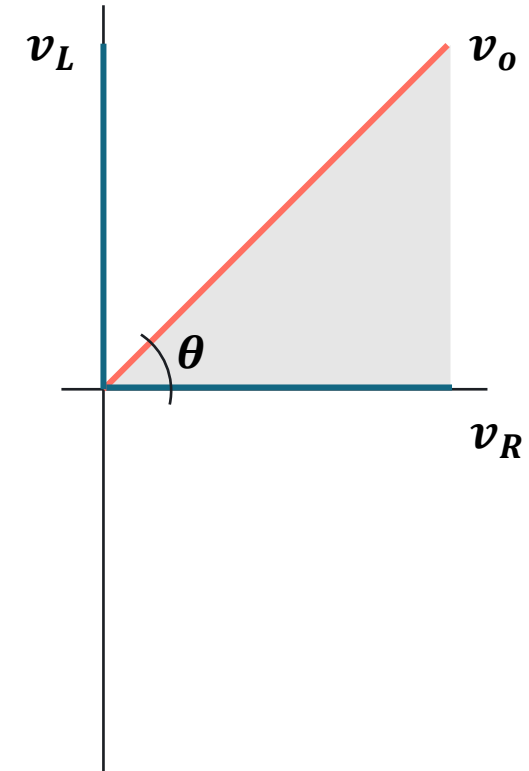
Circuit Diagram



Formula

$$v_o = v_R + jv_L$$

Voltage Phasor Diagram



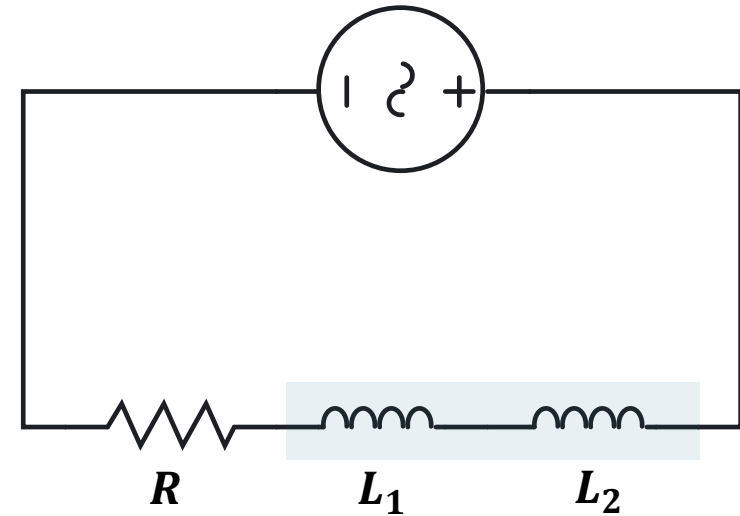
TOTAL INDUCTANCE

Series network

In a series circuit, the total inductance (L_o) is the sum of all individual inductances.

Formula

$$L_o = L_1 + L_2 + L_3 + \cdots L_n$$



EXERCISE

A 240V, 60Hz source is connected to a coil of wire that has a resistance of 7.5Ω and an inductance of $47.7mH$.

Solution

Determine the following:

- a. Equivalent impedance ($\mathbf{Z}$)
- b. Total current ($\mathbf{i_o}$)
- c. Voltage across the resistor ($\mathbf{v_R}$)
- d. Voltage across the inductor ($\mathbf{v_L}$)

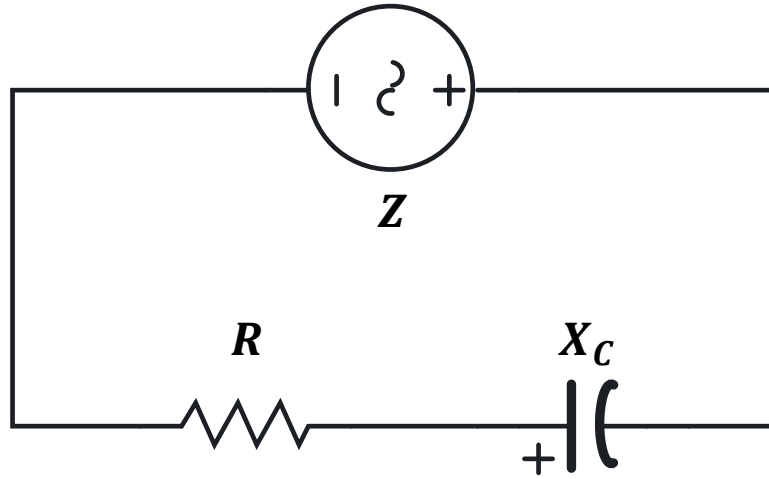


SERIES R-C CIRCUIT



SERIES R-C CIRCUIT

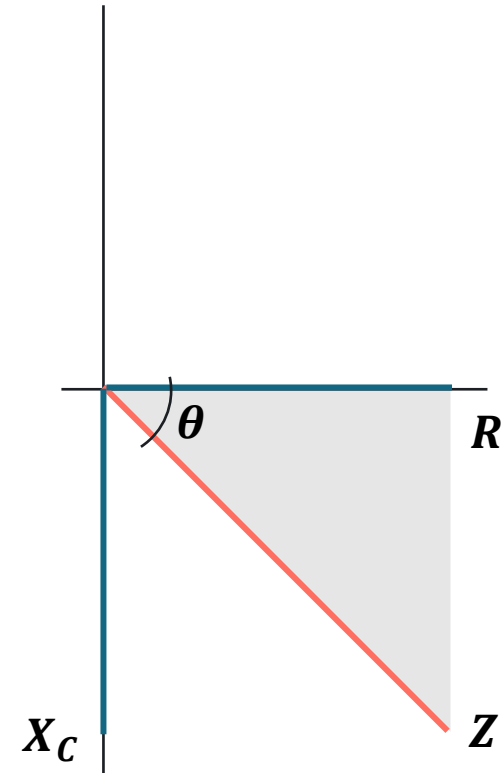
Circuit Diagram



Formula

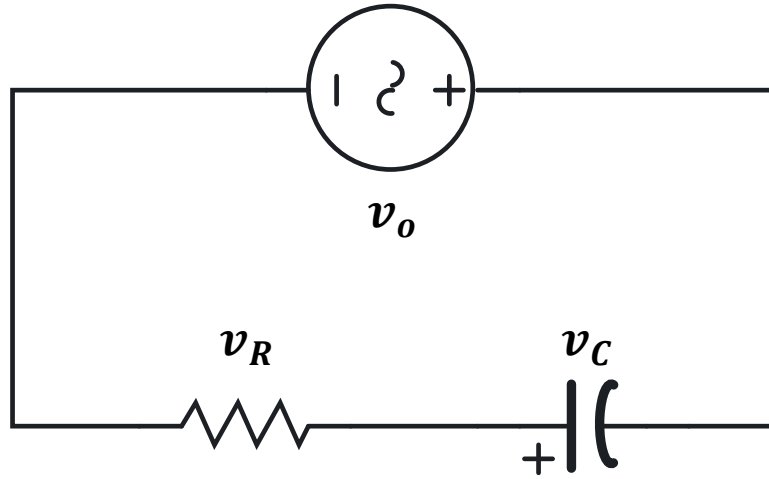
$$Z = R + jX_C$$

Impedance Phasor Diagram



SERIES R-C CIRCUIT

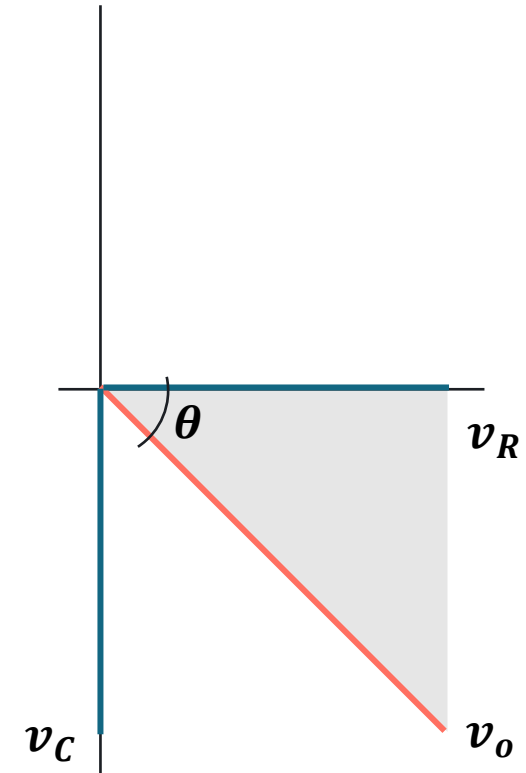
Circuit Diagram



Formula

$$v_o = v_R + jv_C$$

Voltage Phasor Diagram



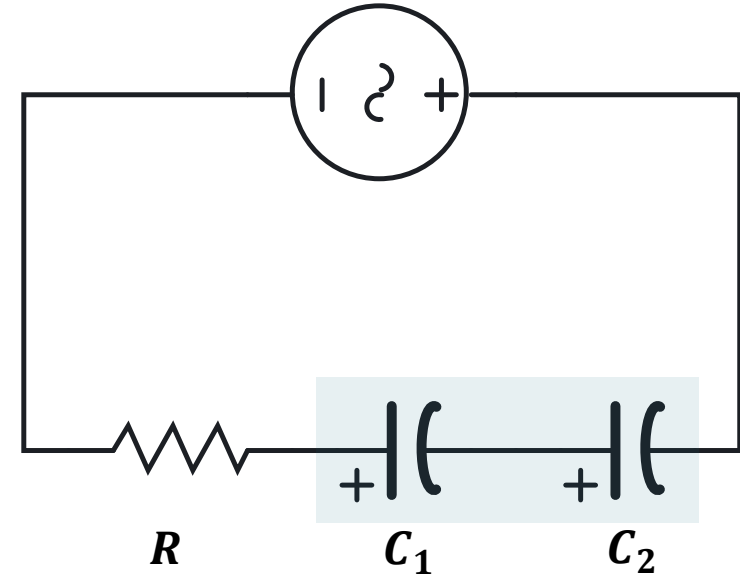
TOTAL CAPACITANCE

In a series circuit, the total capacitance (C_o) is analogous to total resistance in parallel circuit.

Formula

$$\frac{1}{C_o} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$$

Series network



EXERCISE

A 125V, 25Hz source is connected to a series circuit consisting of a $3\ \Omega$ and a $159\mu\text{F}$ capacitor.

Solution

Determine the following:

- a. Equivalent impedance ($\mathbf{Z}$)
- b. Total current ($\mathbf{i_o}$)
- c. Voltage across the resistor ($\mathbf{v_R}$)
- d. Voltage across the capacitor ($\mathbf{v_C}$)

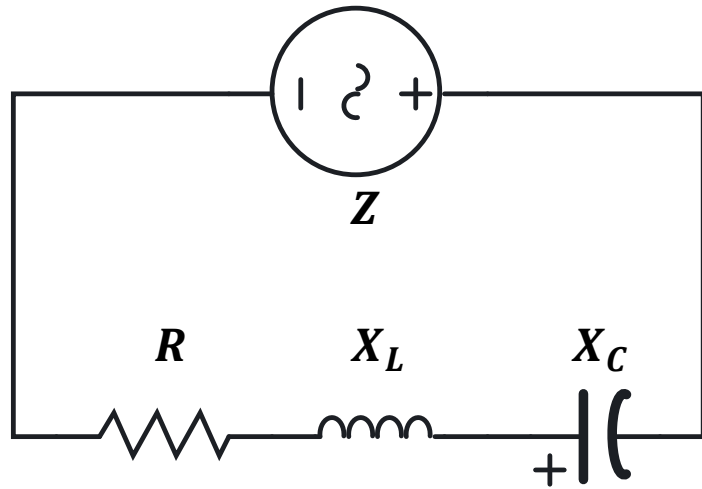


SERIES R-L-C CIRCUIT



SERIES R-L-C CIRCUIT

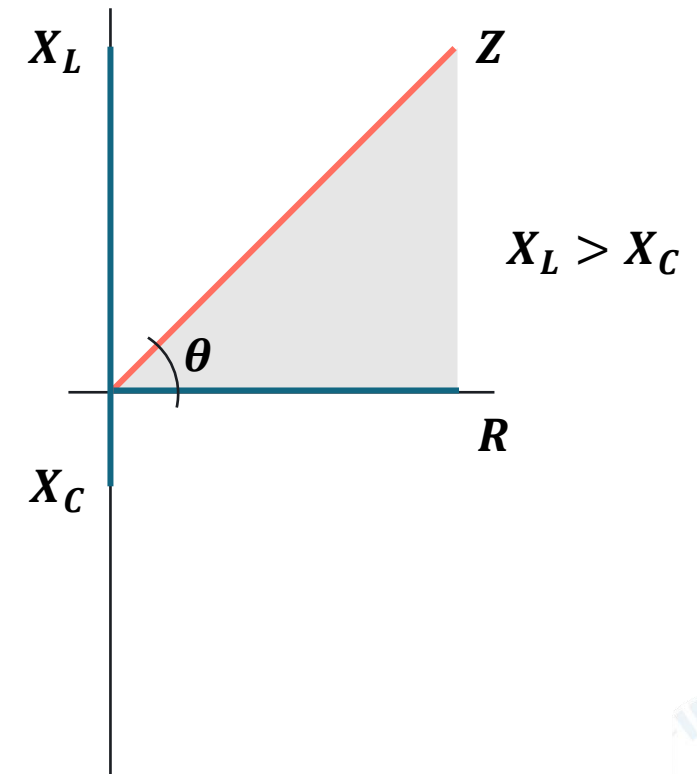
Circuit Diagram



Formula

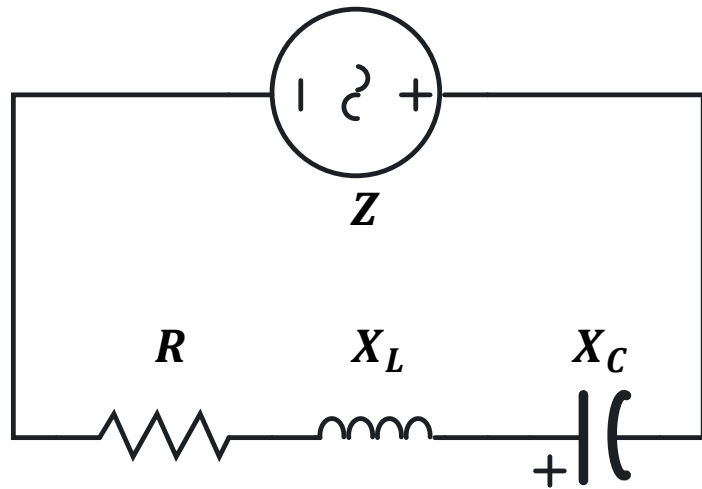
$$Z = R + j(X_L - X_C)$$

Impedance Phasor Diagram



SERIES R-L-C CIRCUIT

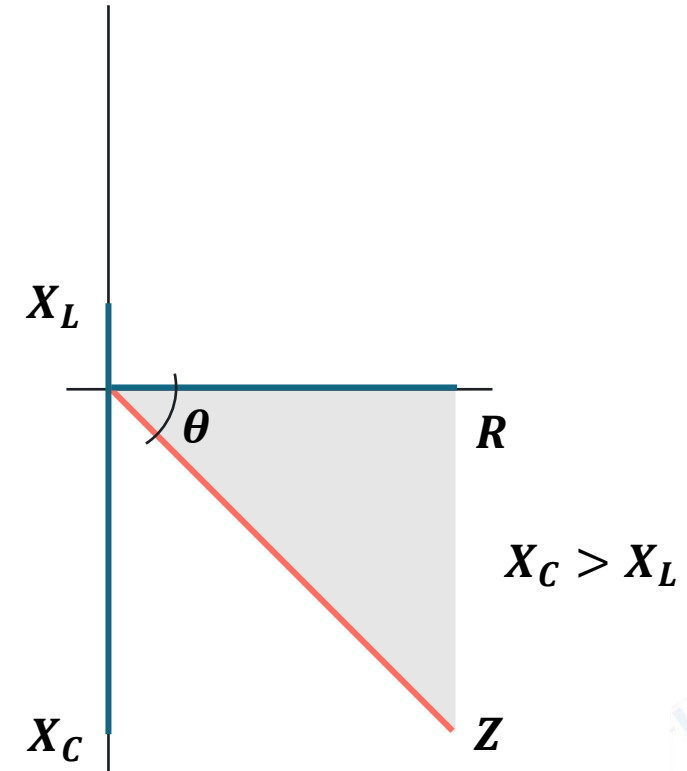
Circuit Diagram



Formula

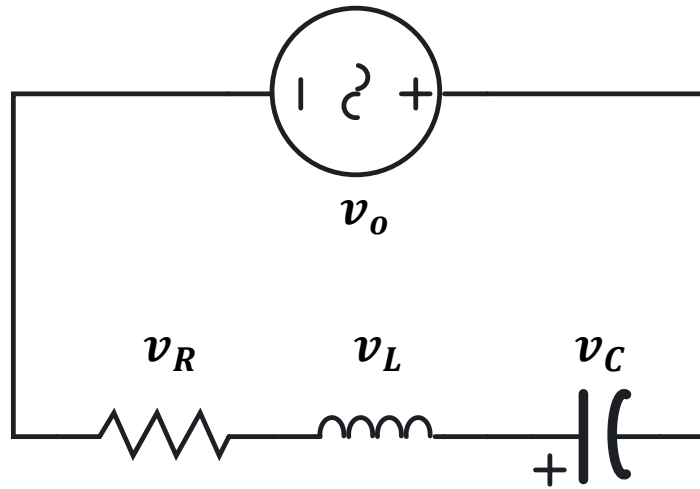
$$Z = R + j(X_L - X_C)$$

Impedance Phasor Diagram



SERIES R-L-C CIRCUIT

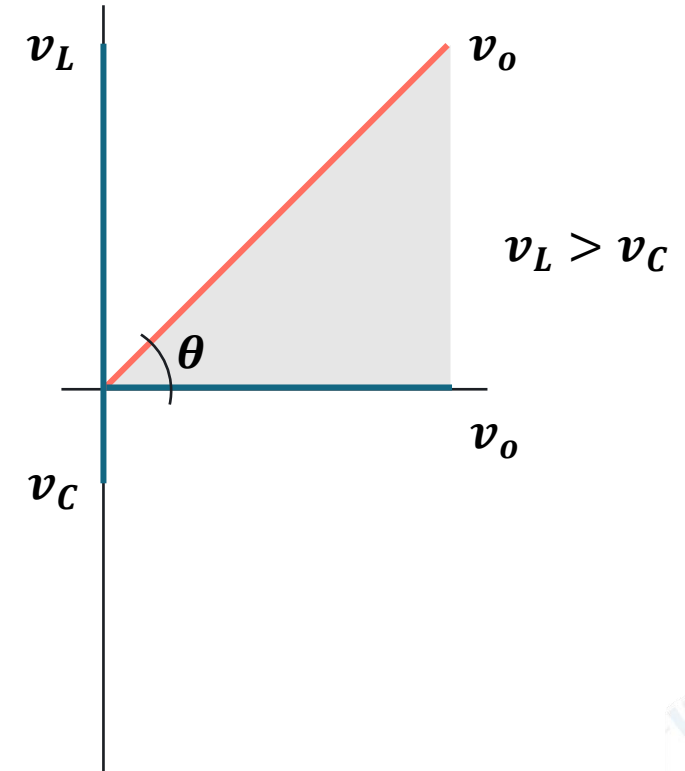
Circuit Diagram



Formula

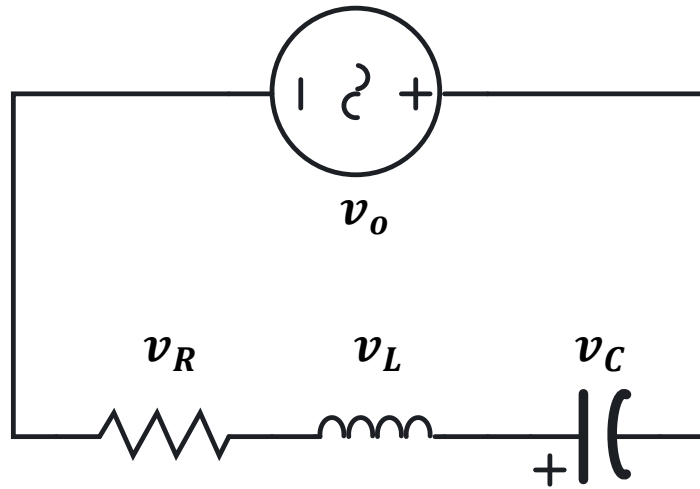
$$v_o = v_R + j(v_L - v_C)$$

Voltage Phasor Diagram



SERIES R-L-C CIRCUIT

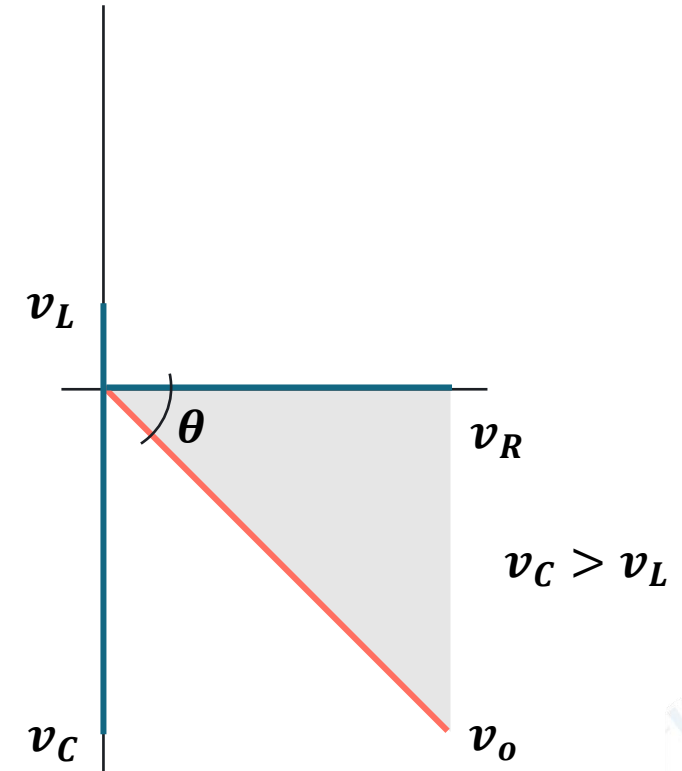
Circuit Diagram



Formula

$$v_o = v_R + j(v_L - v_C)$$

Voltage Phasor Diagram



EXERCISE

A series circuit consisting an 80Ω resistor, a 0.3H inductor, and a $50\mu\text{F}$ capacitor is connected to a 120V , 60Hz source.

Solution

Determine the following:

- Equivalent impedance ($\mathbf{Z}$)
- Total current ($\mathbf{i_o}$)
- Voltage drop across the resistor ($\mathbf{v_R}$)
- Voltage drop across the capacitor ($\mathbf{v_C}$)
- Voltage drop across the inductor ($\mathbf{v_L}$)



POWER FACTOR



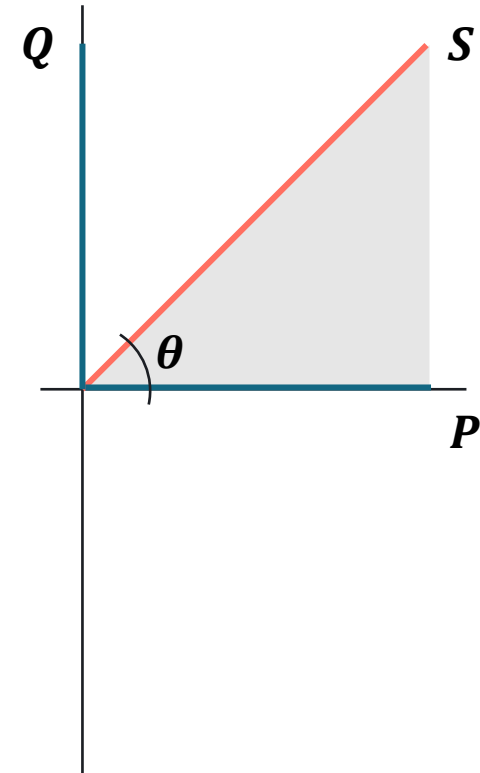
POWER FACTOR

Power Triangle

The power factor ($\cos \theta$) represents the ratio of true power to apparent power.

Formula

$$\cos \theta = \frac{P}{S}$$



TRUE POWER

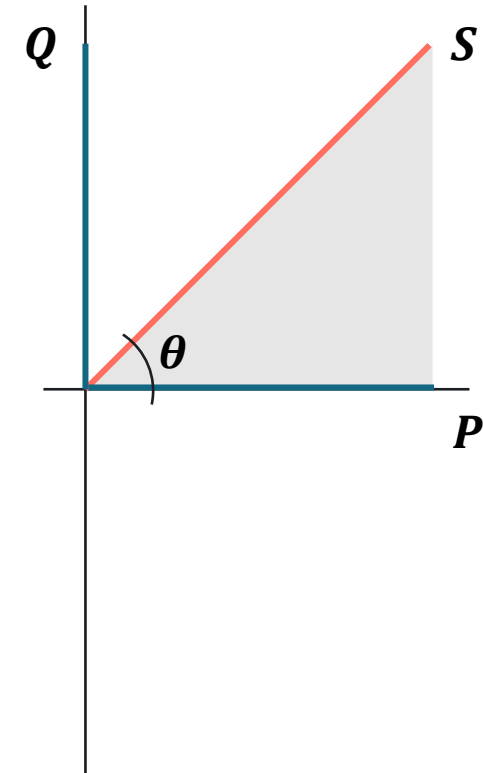
The true power (P) is the actual power consumed by resistive components of a circuit.

Formula

$$P = vi \cos \theta$$

unit: Watt (W)

Power Triangle



REACTIVE POWER

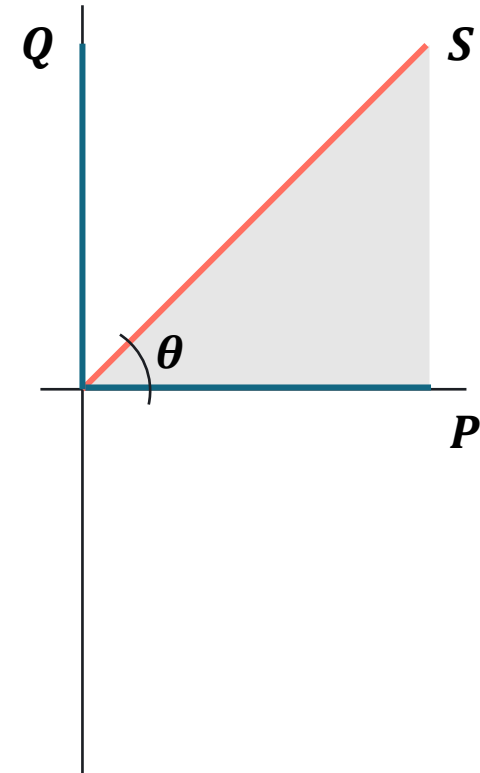
Power Triangle

The reactive power (Q) is the power consumed by inductive or capacitive components of a circuit.

Formula

$$Q = vi \sin \theta$$

unit: Volt-Ampere Reactive (VAR)



APPARENT POWER

The apparent power (S) is the vector sum of true power and reactive power, representing the total power supplied by the source.

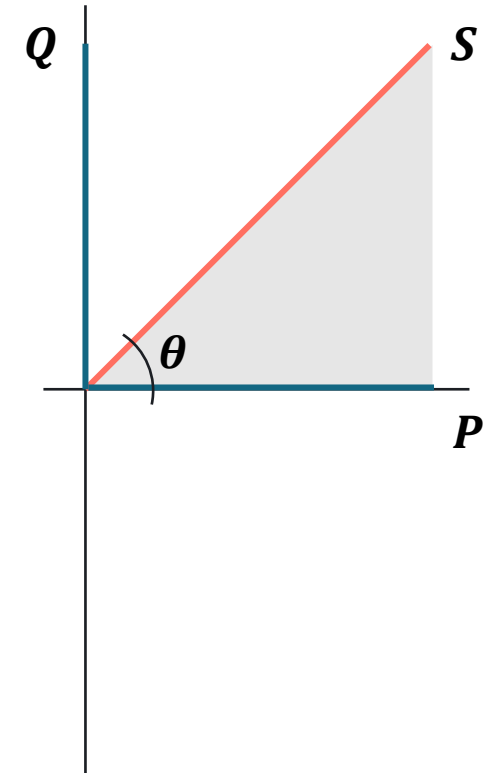
Formula

$$S = P + jQ$$

$$S = vi$$

unit: Volt-Ampere (VA)

Power Triangle



EXERCISE

A series circuit consisting an 80Ω resistor, a 0.3H inductor, and a $50\mu\text{F}$ capacitor is connected to a 120V , 60Hz source.

Solution

Determine the following:

- Power factor ($\cos \theta$)
- True power (P)
- Reactive power (Q)
- Apparent power (S)



LABORATORY

